

Problem 6

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Problem 1 (a) Find the following definite integral:

$$\int_{-4}^4 \frac{x^2 \sin^3(x)}{\sqrt{x^4 + 1}} dx$$

(b) Suppose that f is an even function. Given that $\int_0^6 f(x) dx = 13$, find

$$\int_{-6}^6 (5f(x) + 14) dx.$$